GGS 662: Agentic GeoAI

Course Syllabus, Fall 2026, 3 Credits

Instructor: Ed Oughton

Email: eoughton@gmu.edu

Location: 2310 Exploratory Hall (in-person)

When: Monday 16:30 – 19:10 hrs

Teaching Assistant: N/A

Prerequisites: None (but people without any coding experience may struggle)

Contact method: Canvas discussion board for content related queries (preferred) and/or direct email for anything personal (eoughton@gmu.edu).

OVERVIEW & OBJECTIVES

GGS 662: Agentic GeoAI is a George Mason University course introducing the use of contemporary agentic artificial intelligence (AI) reasoning tools to automate the design, implementation, testing, debugging, refactoring, validation, and documentation of geospatial projects. By the end of the course, students will be able to develop and deploy autonomous reasoning AI agents that can propose, design, execute, evaluate, and iteratively improve traditional GIS routines and workflows.

The course builds on foundational programming and spatial computing skills, emphasizing how AI-assisted reasoning can augment the automation of traditional GIS analysis via AI agents. Particular attention is given to evaluating the correctness and reliability of AI-automated solutions, including the design of testing, verification and validation strategies for GIS codebases, to ensure workflows behave as intended and produce reproducible results. The course also encourages critical reflection on the limitations of AI tool usage in geospatial domains, and the importance of human-in-the-loop decisions.

LEARNING OUTCOMES

The key learning objectives of this class include:

1. Explain the role of AI in geospatial analysis, including how automated workflows utilizing AI reasoning differ from traditional GIS and spatial computing approaches.
2. Use AI reasoning tools effectively to design, write, debug, and refactor geospatial code, demonstrating the ability to translate geographic problems into computational tasks.
3. Integrate AI-generated code with established geospatial libraries and platforms (e.g., Python-based GIS ecosystems) to enable GIS-based scientific discovery.
4. Deploy autonomous AI agents for research, such as to propose, design, build, execute, evaluate and iteratively improve geospatial models and simulations.

5. Critically judge AI-assisted geospatial solutions, including assessing correctness, spatial validity, computational efficiency, and potential sources of error or bias.
6. Develop and implement testing, verification, and validation strategies for GIS-based scientific discovery augmented with AI reasoning.
7. Formulate geographic questions and solve them using AI-enhanced geospatial reasoning, including selecting appropriate methods, tools and data.
8. Reflect on the limitations of AI in geospatial scientific discovery, including issues of transparency, reliability, data quality, and over-reliance on automated solutions.

TIME COMMITMENT

Generally, it is expected that students have some working knowledge of Python. If you do not, then there will be a substantial time commitment. For example, when learning any new language (computer-based or natural), it takes significant effort to progress to fluency. Approximately 10-20 hours of weekly effort is expected outside of class (on assignments, exercises or further reading), with this effort hopefully rewarded later via the job market (as these skills usually translate to higher salaries). If you already have programming experience, this requirement is slightly minimized.

GGS COMPUTER LAB, ASSIGNMENTS, & EXPECTATIONS

GGS students have remote access to the GGS Virtual Computing Lab, although everything for this class should be available via the associated GitHub page (with content going up weekly). Affiliated Google Colab notebooks can be run via a web browser. Thus, all documents, notes and code are available from this single location.

If you do not intend to use the computer lab, then you will need a machine with at least 2 GB RAM, along with a fast and reliable broadband connection (e.g., > 10-20 Mbps). It may also be useful to have a web camera with a microphone in case any additional Zoom sessions are required (or if class is cancelled due to bad weather).

Students will also need to have access to a general piece of GIS software to quickly check/inspect any spatial processing, e.g., via ESRI ArcGIS Pro (licensed) or QGIS (open-source and preferred). Please be aware this is a coding-focused class that will use either of these pieces of software as secondary tools. Students should not expect or seek to use these tools primarily in this class. This is not a class which focuses on GIS analysis using graphical user interfaces (GUIs).

Assignments will be based on the lecture material you receive and will be administered via Canvas. Generally, assignments will be set on a Monday and will be due the following Sunday evening (except when noted in the Course Schedule). Late work will be penalized 20% for each day late. Late submissions will only go unpenalized for documented medical reasons or by previous agreement with the instructor (e.g., raised at the time the assignment is set).

Each student gets the opportunity to drop the two worst performing assignments from the overall score at the end of the semester. Take comfort in the fact that it is highly unlikely that three events justifying extenuating circumstances would occur in a single semester. Thus, the course grading criteria is already accounting for unfortunate events. No additional requests will be accommodated unless the extenuating circumstances are highly serious (in which case the university and course director will already be aware of the issue).

The overall grade comprises three key sets:

GRADING		
Assessment	**Points**	**%** (of final grade)
Assignments (5)	100 each	50%
Coursework project	500	50%

Grading will be based on the following cutoff values, although the instructor reserves the right to alter the values at the end of the course:

A (93%), A- (90%), B+ (87%), B (83%), B- (80%), C+ (77%), C (73%), C- (70%), D (60%)

The coursework project will include the use of techniques taught throughout the entire semester but applied to your own research topic. Students will be expected to submit assignments online through Canvas. Only Adobe PDF (.pdf) file formats will be accepted (students can save documents to a .pdf format for submission).

OPTIONAL TEXTS

There are many open and free resources for learning Python. Importantly, all the information you require will be provided in the course. However, should you require additional information, for example relating to programming languages, there are many options. Check out Al Sweigart's Automate the Boring Stuff with Python which is free to read and provides practical programming for total beginners. You can also check out Ed's YouTube channel for videos on Python-based GIS.

OFFICE HOURS AND INSTRUCTOR INTERACTION

Office hours will be by appointment for longer discussions, or in-person for brief chats prior to each class on Mondays from 16.15 – 16.30 hrs. Each student is expected to attend office hours at least once per semester. When emailing, a timely response is expected during office hours Monday-Friday. If you have a course-related question, the first port of call will be to place it on the Canvas discussion board. This is because other people may ask similar questions, so this becomes a shared knowledge base everyone can access. If you need to speak about something more personal with the instructor, then you can reach out via email. Drop-in virtual Q&A sessions can also be held by request (e.g., Monday,

Tuesday or Wednesday evenings) to seek advice on assignments and coursework.

PROBLEM SOLVING

It is inevitable that problems will arise, especially when working on Python coding problems. Therefore, it is essential that students follow a set of key procedures when dealing with any coding issues encountered. These are as follows:

1. Copy and paste any error messages into a search engine (e.g., Google) or GenAI. Someone else will already have had the same problem, so investigate how other researchers solved similar issues.

2. If you still cannot solve the problem, then consider posting a public question on the Canvas discussion board.

ACADEMIC INTEGRITY

GMU has an Honor Code with clear guidelines regarding academic integrity. Three fundamental and rather simple principles to always follow are that: (1) all work submitted be your own; (2) when using the work or ideas of others, including fellow students, give full credit through accurate citations; and (3) if you are uncertain about the ground rules on a particular assignment, ask for clarification. No grade is important enough to justify academic misconduct (e.g., plagiarism). Another aspect of academic integrity is the free play of ideas. Vigorous discussion and free speech debate are encouraged, with the expectation that all aspects of the class will be conducted with civility and tolerance for differing ideas, perspectives, and traditions.

STUDENT AI USE POLICY

This class recognizes the increasing role that AI tools play in everyday life. Consistent with the GMU AI Guidelines for Students and University academic integrity expectations, you are expected to use AI tools responsibly, transparently, and in ways that support your own learning and the course learning outcomes. Here, you may use AI tools to support your assignments and coursework project, but these tools will not be allowed in the exam. To ensure you comply with GMU policies on this topic, make sure you read and understand the AI guidelines for students: https://www.gmu.edu/ai-guidelines/ai-guidelines-students

Importantly, do not hand in work that looks/reads like AI writing(!). You may think this style is not recognizable, but to experienced people it is often very clear when AI text has been copied verbatim. You absolutely must write your answers (e.g., scientific logic and interpretation/analysis) in your own words. If cheating is suspected, the work will be forwarded to the relevant CoS individual responsible for reviewing code breaches. Interestingly, I am looking for your own take on what you found, even if

that comes with spelling errors and grammatical mistakes (and while these are obviously best avoided), this is far preferable to AI generated text.

INSTRUCTOR AI USE POLICY

This course is compliant with the AI Guidelines for Instructors adapted from the George Mason AI Task Force (as of February 2nd, 2026). All instructional decisions are accountable to the instructor, even when AI supports course content creation. As expected in industry, given this is a computer programming class, Generative AI tools are used to help create code, from possible examples to possible exercises. However, all content is heavily adapted, checked, validated and pre-critiqued by the instructor to ensure a very high quality of delivery, using expert judgement honed over 15 years of experience in the higher education research sector. AI is used to enhance but not replace independent thought. Importantly, integrity is a core part of this class. Therefore, it is expected that students comply with George Mason student AI usage, in the same way it is essential for professors to comply with relevant instructor guidelines.

GMU EMAIL ACCOUNTS

Students must use their Mason email account to receive important University information, including messages related to this class. See http://masonlive.gmu.edu for more information. Please do not email the instructor from a non-GMU email account.

OFFICE OF DISABILITY SERVICES

If you are a student with a disability and you need academic accommodations, please contact the instructor and the Office of Disability Services (ODS) at 993-2474, http://ods.gmu.edu. All academic accommodations must be arranged through the ODS.

GMU RESOURCES

The Writing Center: https://writingcenter.gmu.edu

University Libraries, Ask a Librarian: https://library.gmu.edu/ask

Counseling and Psychological Services: https://caps.gmu.edu

University Catalog: https://catalog.gmu.edu

University Policies: https://universitypolicy.gmu.edu

COURSE OUTLINE

Week	Topic	Coursework
August 24th	Introduction and course overview, what is GeoAI, AI as a coding partner, Python refresh and Colab.	Background reading
August 31st	Agentic problem formulation. Moving from geographic questions to agentic tasks.	Assignment 1
Sept 7th	Labor Day	-
Sept 14th	Agentic route planning application.	-
Sept 21st	Agentic GeoAI applications. Strategies for verification and validation.	Assignment 2
Sept 28th	Agentic AI for scientific discovery in the geosciences.	Assignment 3
Oct 5th	Multi-agent systems for spatial scientific discovery (Part 1).	Assignment 4
Fall Break		
Oct 19th	Multi-agent systems for spatial scientific discovery (Part 2). Isaac Corley, Taylor Geospatial, GeoAI presentation at 5pm.	Assignment 4
Friday Oct 30th	**Special Friday Class** with David Attaway, AWS, 11am-12.30pm. You will each present your coursework project idea.	Assignment 5
Nov 2nd	Agentic GeoAI design and case studies. Caleb Robinson, Microsoft, Agentic GeoAI presentation at 4.30pm.	Coursework Project
Nov 9th	Evaluating agent performance. Justin Guthrie GeoAI presentation on Earth Observation Foundation Models at 5pm.	Coursework Project
Nov 16th	Coursework project.	Coursework Project
Nov 23rd	Coursework project.	Coursework Project
Thanksgiving Recess (Nov 25th-29th)		
Nov 30th	Project presentations and final coursework project audit.	Coursework Project
Dec 7th	Coursework project submission.	-

Note: The course schedule is tentative and is subject to revision by the instructor